Examiner only

5. *(a)* A bottle contains a mixture of liquids **E** and **F**. Liquid **E** has a boiling point of 57 °C and liquid **F** has a boiling point of 95 °C.

Describe the process of distillation and explain why it can be used to separate these liquids. [3]

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(b) One molecule of liquid **E** contains **two** oxygen atoms. The percentage by mass of oxygen in liquid **E** is 43.2 %.

Use the following equation to calculate the relative molecular mass (M_r) of liquid **E**. [2]

$$\frac{\text{mass of oxygen}}{M_r} \times 100 = 43.2$$

$$A_r(\text{O}) = 16$$

M_r = ...

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